Income, Education, and Presidential Voting Patterns Across U.S. Counties

Ben Coleman

Abstract

This project examines how the relationships between county-level income, educational attainment, and presidential voting patterns changed across the 2012, 2016, 2020, and 2024 U.S. presidential elections. County election returns from the MIT Election Data and Science Lab were combined with median household income and bachelor's degree attainment data from the American Community Survey. I used correlations and linear regression to compare the relationships across elections, with 10-fold cross-validation used to evaluate model performance. The results show that educational attainment became increasingly associated with Democratic county vote share, with the correlation increasing from 0.295 in 2012 to 0.552 in 2024. Income also became more positively associated with Democratic vote share, but remained a weaker predictor than education. Models containing both variables performed best overall, although adding income provided only a modest improvement over education alone. These findings suggest that education has become an increasingly important characteristic distinguishing Democratic- and Republican-leaning counties, while the relationship with income is more dependent on its overlap with education.

1 Introduction

Income and education are two informative socioeconomic dimensions of U.S. presidential voting coalitions. Changes in these coalitions can reveal broader shifts in the American electorate, including how the electoral bases of the major political parties have evolved. Previous research has linked income and educational attainment to presidential voting behavior, though these relationships vary by location and election [1, 2, 5]. This project examines how the relationships between county-level income, educational attainment, and presidential vote share have changed across recent U.S. presidential elections.

I investigate these relationships by combining county-level presidential election results with Census and American Community Survey data on median household income and educational attainment across the 2012, 2016, 2020, and 2024 elections. The results show that both socioeconomic characteristics became more strongly associated with presidential vote share over this period, with educational attainment showing the stronger relationship. By comparing these patterns across elections, this analysis provides evidence of a growing educational divide in the socioeconomic geography of presidential support.

2 Related Work

Gelman et al. examined income and presidential voting at the individual, county, and state levels, finding that the relationship between income and Republican support varied substantially across states [2]. Their findings also show that geographic voting patterns do not necessarily reflect individual voting behavior, emphasizing the importance of the level of analysis [2]. Education has also become an increasingly prominent divide in the American electorate. Zingher found that educational attainment became a stronger predictor of partisanship and vote choice after 2000 [5]. At the county level, counties with larger college-educated populations generally shifted toward the Democratic Party between 2000 and 2020, while less-educated counties shifted toward the Republican Party [5]. These socioeconomic relationships have also changed across elections. Feller, Gelman, and Shor found that income-related patterns observed throughout the 2000s did not persist in the same way in 2012, supporting the value of comparing socioeconomic voting patterns over time [1]. Together, this prior research suggests that neither income nor education should be treated as a fixed characteristic of presidential voting coalitions. Their relationships with political support can change across elections and may also depend on the geographic level being examined.

Building on this research, my project examines income and education together at the county level across several recent presidential elections. This allows me to compare whether the relationships identified in prior work remain visible through 2024 and whether one of the two socioeconomic characteristics has become more strongly associated with presidential vote share. Examining both variables in the same models also provides a way to determine whether their relationships with voting patterns remain similar after accounting for their overlap.

3 Data and Methods

3.1 Data Sources

I combined county-level presidential election returns from the MIT Election Data and Science Lab [3] with demographic data from the U.S. Census Bureau's American Community Survey (ACS) [4]. The analysis covers the 2012, 2016, 2020, and 2024 presidential elections. These years provide four election cycles over which changes in the relationships between county socioeconomic characteristics and presidential voting patterns can be compared.

The election data contain candidate vote totals, total votes cast, party, voting mode, and county identifiers. I focused on Democratic and Republican presidential results and calculated each party's share of the total presidential vote within each county. When the election data contained a `TOTAL` voting mode, I used those totals rather than separately summing voting modes. When no total was provided, candidate votes were summed across the available voting modes.

The demographic variables were obtained from the ACS 5-year estimates for each corresponding election year. Median household income was used as the measure of county income. Educational attainment was measured as the percentage of residents age 25 and older with a bachelor's degree or higher. Using the same measures across election years allowed the relationships between these characteristics and presidential vote share to be compared over time.

3.2 Data Cleaning and Integration

The election and ACS datasets were cleaned and merged in R using county FIPS codes and election year. County FIPS codes were standardized to five digits before merging. Differences in geographic reporting created several unmatched observations. In particular, some election records represented geographic units that were not directly comparable to Census counties, including Alaska election districts. A smaller number of observations from other states also could not be reliably matched across the two sources. Rather than attempting to assign demographic characteristics to incompatible geographic units, I excluded observations that could not be matched or were missing one of the variables required for analysis.

After cleaning and merging, the final dataset contained 12,450 county-election observations. This included 3,116 observations for 2012, 3,115 for 2016, 3,114 for 2020, and 3,105 for 2024. The final variables used in the analysis were election year, county identifiers, Democratic vote share, Republican vote share, median household income, and bachelor's degree attainment.

3.3 Analysis

I began with exploratory visualizations and Pearson correlations to examine the direction and strength of the relationships between income, education, and presidential vote share in each election. This provided a direct way to compare how the county-level relationships changed from 2012 through 2024.

Based on the patterns found during exploratory analysis, I then used linear regression to examine the predictive value of the two socioeconomic variables. For each election year, I fit three models predicting Democratic vote share: an income-only model, an education-only model, and a combined model containing both income and education. Comparing the models allowed me to determine whether one characteristic provided substantially more information than the other and whether using both improved model performance.

I also standardized income and education before fitting an additional set of combined models. Because the two variables are measured on very different scales, standardization made their coefficients directly comparable within each election. This analysis was useful for examining how the relationship between each predictor and Democratic vote share changed after accounting for the other predictor.

Finally, I evaluated predictive performance using 10-fold cross-validation. Counties were divided into ten folds within each election year, with each fold used once as the test set while the remaining observations were used for training. Prediction error was measured using root mean squared error (RMSE). This provided an out-of-sample comparison of the income-only, education-only, and combined models rather than relying only on their fit to the same observations used to estimate them.

4 Results

4.1 Changes in Socioeconomic Relationships

The correlation analysis shows that both income and educational attainment became more strongly associated with Democratic county vote share across the four elections (Figure 1). The correlation between educational attainment and Democratic vote share increased from 0.295 in 2012 to 0.428 in 2016, 0.525 in 2020, and 0.552 in 2024. The relationship between median household income and Democratic vote share also became more positive, increasing from 0.087 in 2012 to 0.137 in 2016, 0.268 in 2020, and 0.302 in 2024. Republican vote share showed corresponding negative relationships with both variables.

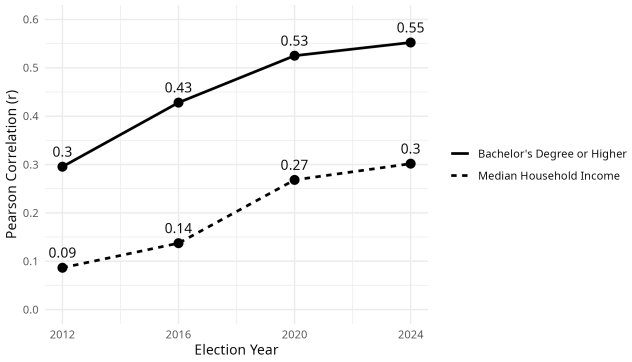


Figure 1: County-level Pearson correlations between median household income, educational attainment, and Democratic presidential vote share across the 2012, 2016, 2020, and 2024 elections.

These results suggest that the socioeconomic geography of presidential voting changed substantially over this period. Education shows the clearest change, with the relationship strengthening in every election examined. Income follows the same general direction, but its relationship with Democratic vote share remains considerably weaker than the relationship with education. Because these are county-level correlations, these patterns describe differences between counties rather than the behavior of individual voters.

4.2 Model Comparison

The regression results provide a similar pattern (Figure 2). Education alone accounted for substantially more variation in Democratic county vote share than income alone in every election. The education-only model's R^2 increased from 0.087 in 2012 to 0.183 in 2016, 0.276 in 2020, and 0.305 in 2024. By comparison, the income-only model increased from an R^2 of 0.007 in 2012 to 0.019 in 2016, 0.072 in 2020, and 0.091 in 2024.

The combined income and education model had the highest R^2 in every election. Its R^2 increased from 0.114 in 2012 to 0.230 in 2016, 0.298 in 2020, and 0.324 in 2024. The improvement from the education-only model to the combined model was relatively small compared with the difference between the income-only and education-only models. This suggests that income contributes some additional information, but education accounts for a larger portion of the variation captured by these two socioeconomic measures.

4.3 Income After Accounting for Education

An additional pattern emerged when income and education were included in the same model using standardized predictors (Figure 3).

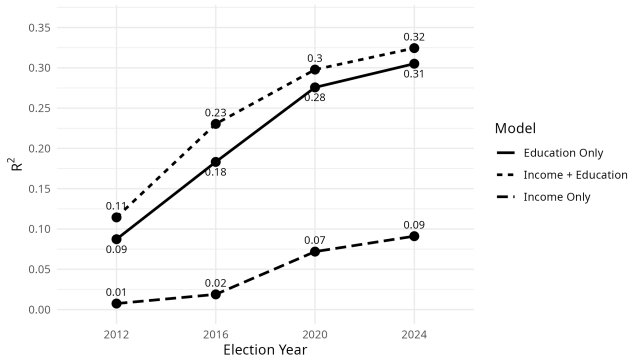


Figure 2: Comparison of model fit across election years using income, educational attainment, and both predictors together. Higher R^2 values indicate that the model accounts for a greater proportion of variation in Democratic county vote share.

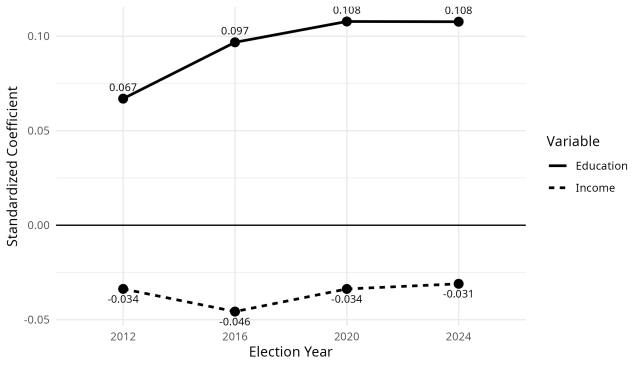


Figure 3: Standardized coefficients for income and educational attainment in the combined models across election years. Positive values indicate a positive association with Democratic county vote share after accounting for the other predictor, while negative values indicate a negative association.

Education maintained a positive coefficient in every election, increasing from 0.067 in 2012 to approximately 0.108 in both 2020 and 2024. In contrast, the standardized income coefficient was slightly negative in every combined model, ranging from approximately −0.031 to −0.046.

This differs from the positive bivariate correlation between income and Democratic vote share. Higher-income counties increasingly tended to have higher Democratic vote shares when income was examined by itself, but the direction changed after educational attainment was included in the model. This suggests that income and education contain overlapping information and that the positive county-level relationship between income and Democratic vote share is partly associated with differences in educational attainment. The regression results do not establish a causal relationship between these characteristics and voting behavior.

Table 1: Mean RMSE from 10-fold cross-validation by model and election year. Lower values indicate better predictive performance.

Model	2012	2016	2020	2024
Income Only	0.147	0.151	0.154	0.147
Education Only	0.141	0.138	0.136	0.129
Income + Education	0.139	0.134	0.134	0.127

5 Evaluation

To evaluate whether the relationships identified by the regression models generalized beyond the observations used to fit them, I used 10-fold cross-validation separately within each election year. Counties were randomly divided into ten folds. Each model was trained on nine folds and evaluated on the remaining fold, with the process repeated until every fold had been used as the test set. I compared the income-only, education-only, and combined income and education models using mean root mean squared error (RMSE), where lower values indicate more accurate predictions of Democratic county vote share.

The cross-validation results were consistent with the model comparisons in the previous section (Table 1). The income-only model had the highest prediction error in every election year, with mean RMSE values of 0.147 in 2012, 0.151 in 2016, 0.154 in 2020, and 0.147 in 2024. The education-only model performed better, with mean RMSE decreasing from 0.141 in 2012 to 0.129 in 2024. The combined model produced the lowest mean RMSE in every election, with values of 0.139, 0.134, 0.134, and 0.127 across the four elections.

The difference between the education-only and combined models was relatively small, particularly in 2012, 2020, and 2024. In contrast, both consistently outperformed the income-only model. This supports the earlier finding that educational attainment provides substantially more information about Democratic county vote share than income alone, while income still provides a modest improvement when the two predictors are used together. The declining RMSE of the education-based models also corresponds with the increasingly strong relationship between educational attainment and Democratic vote share observed across elections.

Cross-validation does not establish that these socioeconomic characteristics cause differences in voting behavior, but it provides a check that the model comparisons are not simply the result of fitting and evaluating the models on the same observations. The consistency between the cross-validation results, correlations, and regression results provides stronger evidence that the changing relationship with education is a meaningful feature of the county-level data.

6 Discussion

The results consistently point to education as the stronger of the two socioeconomic characteristics examined in this project. From 2012 through 2024, counties with higher levels of bachelor's degree attainment became increasingly associated with higher Democratic presidential vote shares. Income moved in the same direction when examined by itself, but its relationship was weaker. The regression and cross-validation results reinforce this distinction, as

education-only models consistently performed substantially better than income-only models.

One result that was less apparent from the initial correlations was the role of income after accounting for education. Although income had a positive bivariate relationship with Democratic vote share, its standardized coefficient became slightly negative when education was included in the same model. This demonstrates why examining the variables together was useful. Income and education capture some overlapping socioeconomic differences between counties, and the positive relationship between income and Democratic vote share cannot be interpreted independently of educational attainment.

The data preparation process also introduced challenges that affected the final analysis. Election results and Census demographic data do not always use directly comparable geographic units, so merging the datasets required more cleaning than I initially expected. Some observations, including Alaska election districts, could not be reliably matched to county-level ACS data and were excluded. The election data also contained different voting modes across states and years, requiring the use of reported total rows when available and aggregation across modes otherwise. Working through these issues made consistent geographic units and aggregation rules an important part of the project rather than simply a preliminary step before modeling.

There are several limitations to the analysis. Most importantly, the results are based on counties rather than individual voters. A relationship between county educational attainment and county Democratic vote share does not demonstrate that individuals with bachelor's degrees voted Democratic at the same rate or that changes in education caused changes in voting behavior. The models also include only two socioeconomic predictors. Other characteristics that vary across counties, such as urbanization, race, age, and regional differences, may account for additional variation and may be related to both education and income.

The project could therefore be improved by incorporating additional demographic and geographic variables and comparing their effects in a broader multivariable model. Population weighting could also provide a different perspective from treating every county equally, since counties vary substantially in population. Finally, extending the analysis to individual-level survey data could help determine whether the county-level patterns found here are also present among individual voters. These additions would provide a stronger basis for separating the role of education and income from other characteristics of the counties in which voters live.

7 Conclusion

This project examined how the relationships between county-level income, educational attainment, and presidential vote share changed across the 2012, 2016, 2020, and 2024 elections. The results show a clear strengthening of the relationship between educational attainment and Democratic vote share. The correlation increased from 0.295 in 2012 to 0.552 in 2024, while education also became substantially more useful for explaining variation in Democratic vote share across counties. Income developed a more positive bivariate relationship with Democratic vote share over the same period, but remained a considerably weaker predictor.

The combined models provided the best overall fit and cross-validated performance, although their improvement over education alone was relatively small. After accounting for education, income also showed a slightly negative relationship with Democratic vote share. Taken together, these findings suggest that the socioeconomic divide between Democratic- and Republican-leaning counties has become increasingly associated with educational attainment, while the role of income is more dependent on its relationship with education.

Future work could expand the analysis by incorporating additional county characteristics such as urbanization, age, race, and region. Comparing population-weighted and unweighted results could also show how the patterns change when larger counties have greater influence on the analysis. Individual-level survey data would provide another useful extension by testing whether the relationships observed between counties are also present among individual voters.

AI Use Disclosure

ChatGPT was used solely to assist in formatting, grammar, and R syntax.

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